

NCERT Exemplar Solutions for Class 12

Chemistry

Chapter 7 – p- Block Elements

I. Multiple Choice Questions (Type-I)

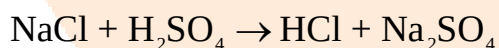
1. On addition of conc. H_2SO_4 to a chloride salt, colourless fumes are evolved but in case of iodide salt, violet fumes come out. This is because

- (i) H_2SO_4 reduces HI to
- (ii) HI is of violet colour
- (iii) HI gets oxidised to I_2
- (iv) HI changes to HI_2O_3

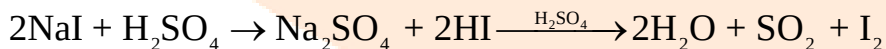
Ans: Correct option is (iii)

Hydrogen iodide is a more powerful reducing agent than sulphuric acid, it reduces the amount of iodine in the solution from H_2SO_4 to SO_2 and HI to I_2

When chloride salts are treated with sulfuric acid, HCl gas is formed, which produces a colourless gas.



Violet fumes are produced during the reaction due to the creation of iodine (I_2) gas.



2. In qualitative analysis when H_2S is passed through an aqueous solution of salt acidified with dil. HCl, a black precipitate is obtained. On boiling the

precipitate with dil. HNO_3 , it forms a solution of blue colour. Addition of excess of aqueous solution of ammonia to this solution gives _____.

- (i) deep blue precipitate of $\text{Cu}(\text{OH})_2$
- (ii) deep blue solution of $[\text{Cu}(\text{NH}_3)_4]^{2+}$
- (iii) deep blue solution of $\text{Cu}(\text{NO}_3)_2$
- (iv) deep blue solution of $\text{Cu}(\text{OH})_2 \cdot \text{Cu}(\text{NO}_3)_2$

Ans: Correct option (ii)

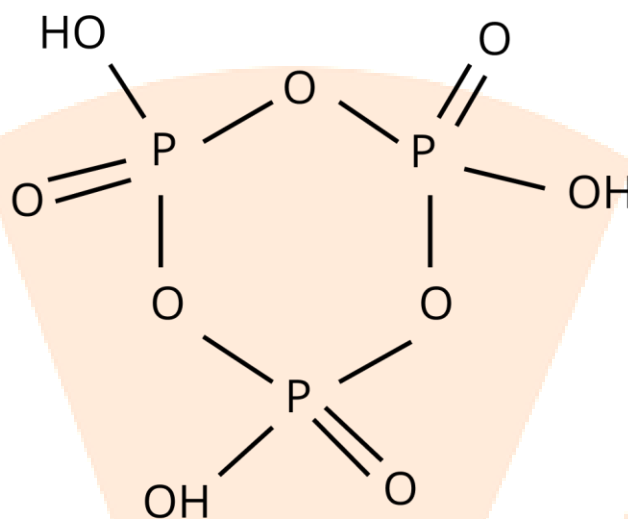
When H_2S is passed through an aqueous solution of salt acidified with dil. HCl in qualitative analysis, a black precipitate is produced. When the precipitate is boiled with dil. HNO_3 , a blue solution results. When an excess of ammonia aqueous solution is added to this solution, it produces a deep blue $[\text{Cu}(\text{NH}_3)_4]^{2+}$ solution.

3. In a cyclotrimetaphosphoric acid molecule, how many single and double bonds are present?

- (i) 3 double bonds; 9 single bonds
- (ii) 6 double bonds; 6 single bonds
- (iii) 3 double bonds; 12 single bonds
- (iv) Zero double bonds; 12 single bonds

Ans: Correct option is (iii)

H_3PO_2 , H_3PO_3 , H_3PO_4 , and other oxoacids are formed by phosphorus. P is tetrahedrally surrounded by other atoms in phosphorus oxoacids. At least one P-H bond and an O-H bond are known to form in all of them.



4. Which of the following elements can be involved in $p\pi - d\pi$ bonding?

(i) Carbon

(ii) Nitrogen

(iii) Phosphorus

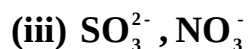
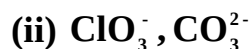
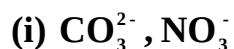
(iv) Boron

Ans: Correct option (iii)

If an electron pair is donated to a vacant orbital on one atom and a lone pair of electrons on the other, the bonding is designated $p\pi - p\pi$ depending on the orbital to which the electron pair is donated and the orbital from which the electron pair is donated.

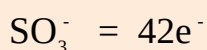
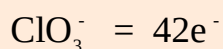
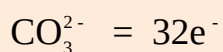
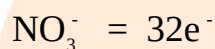
Because their valence shells lack d-orbitals, nitrogen, carbon, and boron cannot form a pi bond. Phosphorus, on the other hand, can produce pi bonds.

5. Which of the following pairs of ions are isoelectronic and isostructural?



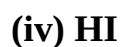
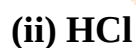
Ans: Correct option (i)

The compound has the following number of electrons:



Hence, NO_3^- and CO_3^{2-} are isoelectronic.

6. Affinity for hydrogen decreases in the group from fluorine to iodine. Which of the halogen acids should have highest bond dissociation enthalpy?



Ans: Correct option (i)

Fluorine is the most reactive element, forming hydrogen fluoride when it combines with hydrogen gas.

Due to the shorter bond length between the two atoms, it takes a lot of energy to break the link between hydrogen and fluorine.

As a result, as we move down the group, the bond dissociation energy drops.

7. Bond dissociation enthalpy of E-H (E = element) bonds is given below. Which of the compounds will act as the strongest reducing agent?

Compound	NH ₃	PH ₃	AsH ₃	SbH ₃
$\Delta_{\text{diss}}(\text{E} - \text{H}) / \text{kJ mol}^{-1}$	389	322	297	255

(i) NH₃

(ii) PH₃

(iii) AsH₃

(iv) SbH₃

Ans: Correct option is (iv)

The E-H bond dissociation energy of SbH₃ is the lowest among the aforementioned compounds, it easily releases H, making it a strong reducing agent. The greater the lowering agent, the weaker the E-H bond.

8. On heating with concentrated NaOH solution in an inert atmosphere of CO₂, white phosphorus gives a gas. Which of the following statements is incorrect about the gas?

(i) It is highly poisonous and smells like rotten fish.

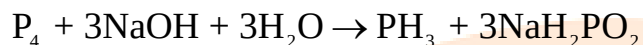
(ii) It's solution in water decomposes in the presence of light.

(iii) It is more basic than NH₃.

(iv) It is less basic than NH₃.

Ans: Correct option is (iii)

When heated with concentrated NaOH in an inert CO_2 environment. It forms PH_3



As we progress from the top to the bottom in a group, the basic character of hydrides reduces PH_3 is basic than NH_3 .

9. Which of the following acids forms three series of salts?

(i) H_3PO_2

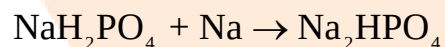
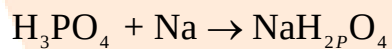
(ii) H_3BO_3

(iii) H_3PO_4

(iv) H_3PO_3

Ans: Correct option is (iii) H_3PO_4

When it reacts with a metal atom, it can produce three different forms of salts, referred known as the three series of salts.



10. Strong reducing behaviour of H_3PO_2 is due to

(i) Low oxidation state of phosphorus

(ii) Presence of two $-\text{OH}$ groups and one $\text{P}-\text{H}$ bond

(iii) Presence of one $-\text{OH}$ group and two $\text{P}-\text{H}$ bonds

(iv) High electron gain enthalpy of phosphorus

Ans: Correct option is (iii)

H_3PO_2 has one O-H group and two P-H bonds

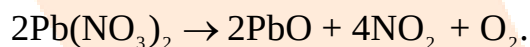
The existence of a P-H link gives phosphorus oxyacids their reducing characteristics. It has a strong proclivity for releasing protons. As a result, it demonstrates a decreasing nature.

11. On heating lead nitrate forms oxides of nitrogen and lead. The oxides formed are _____.

- (i) N_2O , PbO
- (ii) NO_2 , PbO
- (iii) NO , PbO
- (iv) NO , PbO_2

Ans: Correct option is (ii)

On heating, lead nitrate decomposes into lead monoxide, nitrogen dioxide, and oxygen.



12. Which of the following elements does not show allotropy?

- (i) Nitrogen
- (ii) Bismuth
- (iii) Antimony
- (iv) Arsenic

Ans: Correct option is (i)

The single N-N bond is weak because of high inter-electronic repulsion of the non-bonding electrons, owing to the small bond length. As a result the catenation tendency is weaker in nitrogen that is why it does not show allotropy.

13. Maximum covalency of nitrogen is _____.

- (i) 3
- (ii) 5
- (iii) 4
- (iv) 6

Ans: Correct option is (ii)

The valence electrons of nitrogen are two 2s and three 2p. Nitrogen can make three covalent bonds by sharing its three 2p electrons. Consider the NH_3 molecule. The nitrogen atom, however, still contains a lone pair of electrons in the 2s orbital. It can make one bond by transferring these two electrons from the lone pair. Consider the NH_4^+ molecule.

As a result, nitrogen can create four different bonds.

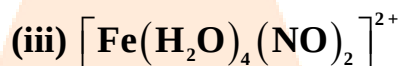
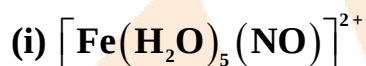
14. Which of the following statements is wrong?

- (i) Single N–N bond is stronger than the single P–P bond.
- (ii) PH_3 can act as a ligand in the formation of coordination compound with transition elements.
- (iii) NO_2 is paramagnetic in nature.
- (iv) Covalency of nitrogen in N_2O_5 is four

Ans: Correct option is (i)

The N-N sigma bond (single bond) is weaker than the P-P sigma bond (single bond) due to the smaller size of nitrogen which increases the interelectronic repulsions and thus making the bond weaker.

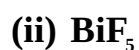
15. A brown ring is formed in the ring test for NO_3^- ion. It is due to the formation of



Ans: Correct option is (i)

The 'Brown Ring Test' is an identification and conformation test of the nitrate ion (NO_3^-) using conc. H_2SO_4 and newly produced FeSO_4 . Place the NO_3^- containing sample in a test tube. Then, with gentle shaking, add conc H_2SO_4 dropwise, heat it briefly on a Bunsen burner, and add freshly made greenish FeSO_4 . Then, due to the creation of brown coloured $[\text{Fe}(\text{H}_2\text{O})_5(\text{NO})]\text{SO}_4$ complex, a brown ring forms in that test tube.

16. Elements of group-15 form compounds in +5 oxidation state. However, bismuth forms only one well characterised compound in +5 oxidation state. The compound is



(iv) Bi_2S_5

Ans: Correct option is (ii)

Bismuth is the sole element that has an inert pair effect. Only generates trihalides and has a +3 oxidation state. Fluorine, on the other hand, is small and has a high electronegativity.

17. On heating ammonium dichromate and barium azide separately we get

(i) N_2 in both cases

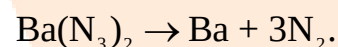
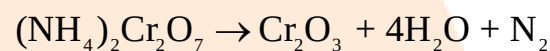
(ii) N_2 with ammonium dichromate and NO with barium azide

(iii) N_2O with ammonium dichromate and N_2 with barium azide

(iv) N_2O with ammonium dichromate and NO_2 with barium azide

Ans: Correct option is (i)

We get N_2 in both situations when we heat ammonium dichromate and barium azide separately.



18. In the preparation of HNO_3 , we get NO gas by catalytic oxidation of ammonia. The moles of NO produced by the oxidation of two moles of NH_3 will be _____.

(i) 2

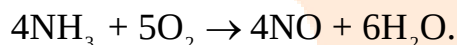
(ii) 3

(iii) 4

(iv) 6

Ans: Correct option is (i)

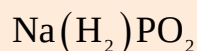
The catalytic oxidation of ammonia produces NO gas, which is used to make HNO_3 . 4 moles of NH_3 created 4 moles of NO in the equation below. As a result, the moles of NO produced by oxidising two moles of NH_3 will be two moles.



19. The oxidation state of central atom in the anion of compound NaH_2PO_2 will be _____.

- (i) +3
- (ii) +5
- (iii) +1
- (iv) -3

Ans: Correct option is (iii)



$$= +1 + (2x + 1) + x + 2(-2) = 0$$

$$x - 1 = 0$$

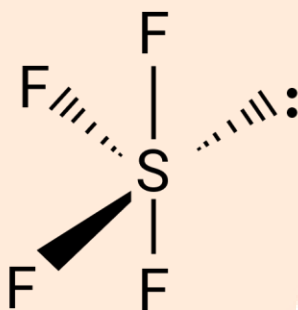
$$\text{or } x = 1$$

20. Which of the following is not tetrahedral in shape?

- (i) NH_4^+
- (ii) SiCl_4
- (iii) SF_4
- (iv) SO_4^{2-}

Ans: Correct option is (iii)

The C-atom in the CO_3^{2-} ion undergoes sp^2 hybridization. BF_4 , NH_4^+ and SO_4^{2-} have a tetrahedral structure, whereas it has a triangular planar structure.



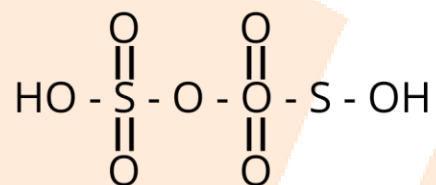
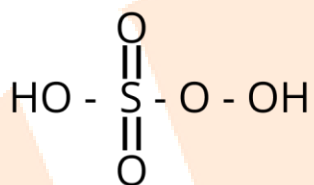
21. Which of the following are peroxyacids of sulphur?

- (i) H_2SO_5 and $\text{H}_2\text{S}_2\text{O}_8$
- (ii) H_2SO_5 and $\text{H}_2\text{S}_2\text{O}_7$
- (iii) $\text{H}_2\text{S}_2\text{O}_7$ and $\text{H}_2\text{S}_2\text{O}_8$
- (iv) $\text{H}_2\text{S}_2\text{O}_6$ and $\text{H}_2\text{S}_2\text{O}_7$

Ans: Correct option is (i)

Peroxy linkage refers to the presence of a bond between oxygen and oxygen (O-O) in a molecule.

Only $\text{H}_2\text{S}_2\text{O}_6$ and $\text{H}_2\text{S}_2\text{O}_7$ of the sulphur oxoacids above have a speroxy linkage.



22. Hot conc. H_2SO_4 acts as moderately strong oxidising agent. It oxidises both metals and nonmetals. Which of the following element is oxidised by conc. H_2SO_4 into two gaseous products?

- (i) Cu
- (ii) S
- (iii) C
- (iv) Zn

Ans: Correct option is (iii)



Conc. H_2SO_4 oxidises C to produce two gaseous products.

(i) covalent solid

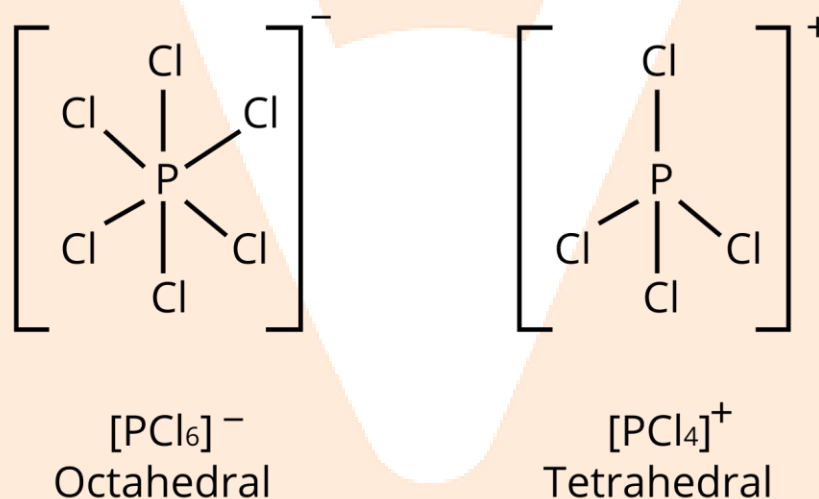
(ii) octahedral structure

(iii) ionic solid with $[\text{PCl}_6]^+$ octahedral and $[\text{PCl}_4]^-$ tetrahedra

(iv) ionic solid with $[\text{PCl}_4]^+$ tetrahedral and $[\text{PCl}_6]^-$ octahedra

Ans: Correct option is (iv)

The ionic bonding promotes the crystalline structure of PCl_5 in the solid state by attempting to exist as oppositely charged ions like $[\text{PCl}_4]^+$ and $[\text{PCl}_6]^-$. Also, $[\text{PCl}_4]^+$ and $[\text{PCl}_6]^-$ are tetrahedral and octahedral, respectively. These structures fit together well, giving the solid structure extra stability.



26. Reduction potentials of some ions are given below. Arrange them in decreasing order of oxidising power.

Ion	ClO_4^-	IO_4^-	BrO_4^-
Reduction potential E^- / V	$E^- = 1.19 \text{ V}$	$E^- = 1.74 \text{ V}$	$E^- = 1.74 \text{ V}$

(i) $\text{ClO}_4^- > \text{IO}_4^- > \text{BrO}_4^-$ –

(ii) $\text{IO}_4^- > \text{BrO}_4^- > \text{ClO}_4^-$

(iii) $\text{BrO}_4^- > \text{IO}_4^- > \text{ClO}_4^-$

(iv) $\text{BrO}_4^- > \text{ClO}_4^- > \text{IO}_4^-$

Ans: Correct option is (iii)

The ability of a substance to be decreased is known as its reduction potential. As the substance's reduction potential rises, so does its power to reduce. It signifies the chemical's oxidising power (the ability of the material to cause other substances to lose electrons, which is known as oxidising power) grows.

As a result, the oxidising power is listed in decreasing order $\text{BrO}_4^- > \text{IO}_4^- > \text{ClO}_4^-$

27. Which of the following is isoelectronic pair?

(i) ICl_2 , ClO_2

(ii) BrO_2^- , BrF_2

(iii) ClO_2 , BrF

(iv) CN^- , O_3

Ans: Correct option is (ii)

BrO_2^- ($35 + 2 \times 8 + 1 = 52$) and BrF_2^+ ($35 + 2 \times 9 - 1 = 52$)

52 electrons are present in both BrO_2^- , BrF_2

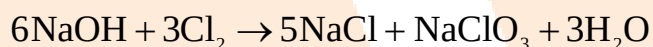
II. Multiple Choice Questions (Type-II)

Note: In the following questions two or more options may be correct.

28. If chlorine gas is passed through hot NaOH solution, two changes are observed in the oxidation number of chlorine during the reaction. These are _____ and _____.

- (i) 0 to +5
- (ii) 0 to +3
- (iii) 0 to -1
- (iv) 0 to +1

Ans: Correct option is (i) and (iii)



Chlorine gas oxidation number ranges from 0 to -1 to 0 to +5.

29. Which of the following options are not in accordance with the property mentioned against them?

- (i) $\text{F}_2 > \text{Cl}_2 > \text{Br}_2 > \text{I}_2$ Oxidising power.
- (ii) $\text{MI} > \text{MBr} > \text{MCl} > \text{MF}$ Ionic character of metal halide.
- (iii) $\text{F}_2 > \text{Cl}_2 > \text{Br}_2 > \text{I}_2$ Bond dissociation enthalpy.
- (iv) $\text{HI} < \text{HBr} < \text{HCl} < \text{HF}$ Hydrogen-halogen bond strength.

Ans: Correct option is (ii), (iii)

$\text{MI} > \text{MBr} > \text{MCl} > \text{MF}$ is an ionic character of metal halide

$\text{F}_2 > \text{Cl}_2 > \text{Br}_2 > \text{I}_2$ is a bond dissociation enthalpy

30. Which of the following is correct for P_4 molecule of white phosphorus?

- (i) It has 6 lone pairs of electrons.**
- (ii) It has six P–P single bonds.**
- (iii) It has three P–P single bonds.**
- (iv) It has four lone pairs of electrons.**

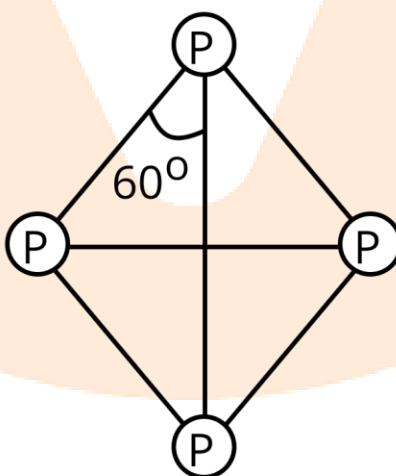
Ans: Correct option is (ii) and (iv)

Tetrahedral geometry characterises the structure of a white phosphorus molecule.

Phosphorus has a valency of 5, and there are four phosphorus atoms in total. As a result, there will be a total of $4 \times 5 = 20$ valence electrons.

As a result, each P atom will have one lone pair of electrons, and each covalent bond (two electrons) will be established.

In a molecule of white phosphorus, there will be four lone pairs of electrons and six P-P single bonds in total.



31. Which of the following statements are correct?

- (i) Among halogens, radius ratio between iodine and fluorine is maximum.**
- (ii) Leaving F—F bond, all halogens have weaker X—X bond than X—X' bond in interhalogens.**
- (iii) Among interhalogen compounds maximum number of atoms are present in iodine fluoride.**
- (iv) Interhalogen compounds are more reactive than halogen compounds.**

Ans: Correct option is (i)

Due to its high electronegativity and tiny size, F is the only interhalogen with weaker X-X' bonds and no X-X bonds.

32. Which of the following statements are correct for SO₂ gas?

- (i) It acts as bleaching agent in moist conditions.**
- (ii) It's molecule has linear geometry.**
- (iii) It's dilute solution is used as disinfectant.**
- (iv) It can be prepared by the reaction of dilute H₂SO₄ with metal sulphide.**

Ans: Correct option is (i) and (iii)

SO₂ is employed as an antichlor, disinfectant, and preservative in the bleaching of wool and silk.

33. Which of the following statements are correct?

- (i) All the three N—O bond lengths in HNO₃ are equal.**

(ii) All P—Cl bond lengths in PCl_5 molecule in gaseous state are equal.

(iii) P_4 molecule in white phosphorus have angular strain therefore white phosphorus is very reactive.

(iv) PCl is ionic in solid state in which cation is tetrahedral and anion is octahedral.

Ans: Correct option is (iii) and (iv)

The P_4 molecule in white phosphorous has an angular strain, white phosphorous is very reactive.

PCl_5 is ionic in the solid state, with a tetrahedral cation and an octahedral anion.

34. Which of the following orders are correct as per the properties mentioned against each?

(i) $\text{As}_2\text{O}_3 < \text{SiO}_2 < \text{P}_2\text{O}_3 < \text{SO}_2$ **Acid strength.**

(ii) $\text{AsH}_3 < \text{PH}_3 < \text{NH}_3$ **Enthalpy of vapourisation.**

(iii) $\text{S} < \text{O} < \text{Cl} < \text{F}$ **More negative electron gain enthalpy.**

(iv) $\text{H}_2\text{O} > \text{H}_2\text{S} > \text{H}_2\text{Se} > \text{H}_2\text{Te}$ **Thermal stability.**

Ans: Correct option is (i) and (iv)

(i) $\text{As}_2\text{O}_3 < \text{SiO}_2 < \text{P}_2\text{O}_3 < \text{SO}_2$ is the order of acid

(ii) $\text{AsH}_3 < \text{PH}_3 < \text{NH}_3$ is the correct order of enthalpy of vapourization

(iii) $\text{S} < \text{O} < \text{F} < \text{Cl}$ is the correct order of more negative gain enthalpy

(iv) $\text{H}_2\text{O} > \text{H}_2\text{Se} > \text{H}_2\text{Te}$ is the order of thermal stability

35. Which of the following statements are correct?

(i) S–S bond is present in $\text{H}_2\text{S}_2\text{O}_6$.

(ii) In peroxosulphuric acid H_2SO_5 sulphur is in +6 oxidation state.

(iii) Iron powder along with Al_2O_3 and K_2O is used as a catalyst in the preparation of NH_3 by Haber's process.

(iv) Change in enthalpy is positive for the preparation of SO_3 by catalytic oxidation of SO_2

Ans: Correct option is (i)

Oxidation state of S = +6

(iii) Iron oxide with K_2O and Al_2O_3 is used to increase the rate of attainment of equilibrium in Haber's process.

(iv) Change in enthalpy is negative for the preparation of SO_3 by catalytic oxidation of SO_2 .

36. In which of the following reactions conc. H_2SO_4 is used as an oxidising reagent?

(i) $\text{CaF}_2 + \text{H}_2\text{SO}_4 \rightarrow \text{CaSO}_4 + 2\text{HF}$

(ii) $2\text{HI} + \text{H}_2\text{SO}_4 \rightarrow \text{I}_2 + \text{SO}_2 + 2\text{H}_2\text{O}$

(iii) $\text{Cu} + 2\text{H}_2\text{SO}_4 \rightarrow \text{CuSO}_4 + \text{SO}_2 + 2\text{H}_2\text{O}$

(iv) $\text{NaCl} + \text{H}_2\text{SO}_4 \rightarrow \text{NaHSO}_4 + \text{HCl}$

Ans: Correct option is (ii) and (iii)

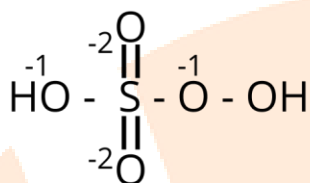


Fig. A

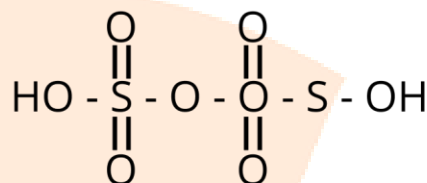


Fig. B

Here fig (b) contains one S-S bond.

The oxidising behaviour of H_2SO_4 is represented by (ii) and (iii) among the aforementioned four. It oxidises HI in reaction (ii) and then reduces to SO_2 .

The oxidation state of sulphur's core atom drops from +6 to +4. It oxidises copper in (iii) and is reduced to SO_2 .

37. Which of the following statements are true?

- (i) Only type of interactions between particles of noble gases are due to weak dispersion forces.**
- (ii) Ionisation enthalpy of molecular oxygen is very close to that of xenon.**
- (iii) Hydrolysis of XeF_6 is a redox reaction.**
- (iv) Xenon fluorides are not reactive.**

Ans: Correct option is (i) and (ii)

(i) The modest dispersion force causes attraction in noble gases.

(ii) In nature, xenon fluorides are reactive.

III. Short Answer Type Question

38. In the preparation of H_2SO_4 by Contact Process, why is SO_3 not absorbed directly in water to form H_2SO_4 ?

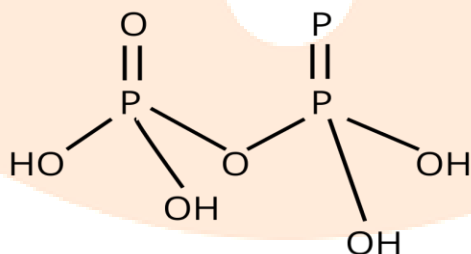
Ans: SO_3 generates a dense fog of sulphuric acid that does not condense quickly, it is not absorbed directly in water to form H_2SO_4 .

39. Write a balanced chemical equation for the reaction showing catalytic oxidation of NH_3 by atmospheric oxygen.

Ans: $4\text{NH}_3 + 5\text{O}_2 \xrightarrow[500\text{k}, 9\text{bar}]{\text{pt/Raue catalyst}} 4\text{NO} + 6\text{H}_2\text{O}$

40. Write the structure of pyrophosphoric acid.

Ans:

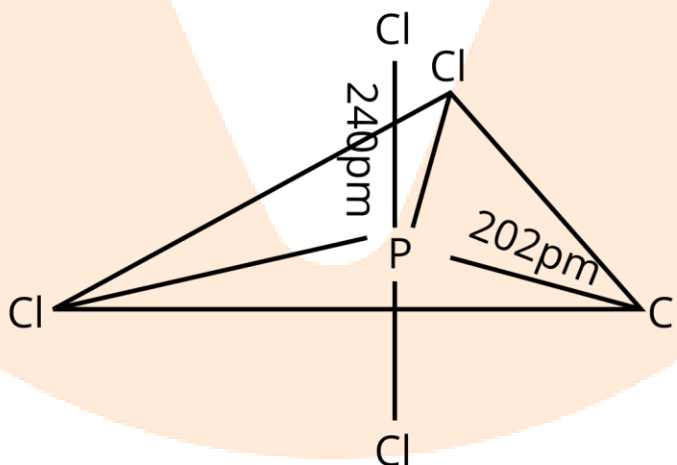


41. PH_3 forms bubbles when passed slowly in water but NH_3 dissolves. Explain why?

Ans: PH_3 is insoluble in water and cannot create hydrogen bonds with it, it forms bubbles, whereas NH_3 dissolves because it is soluble in water and can form hydrogen bonds with it.

42. In PCl_5 , phosphorus is in sp^3d hybridised state but all its five bonds are not equivalent. Justify your Answer with reason.

Ans: Three P—Cl bonds are arranged in one plane at a 120° angle. These bonds are known as equatorial bonds because they are formed between two points. The remaining two P—Cl bonds, one above the equatorial plane and the other below it, form 90° angle with the plane. Axial bonds are the name for these types of bonds. Because axial bond pairs are subjected to greater repulsive interaction than equatorial bond pairs, axial bonds are slightly longer and hence slightly weaker than equatorial bonds, making the PCl_5 molecule more reactive.



43. Why is nitric oxide paramagnetic in gaseous state but the solid obtained on cooling it is diamagnetic?

Ans: NO_2 exists as a monomer with one unpaired electron in the gaseous state, but it dimerises to N_2O_4 in the solid state, leaving no unpaired electron, making the solid form diamagnetic.

44. Give reason to explain why ClF_3 exists but FCl_3 does not exist.

Ans: Chlorine has vacant d-orbitals, which become excited upon bonding when electrons from the 3p-orbital are promoted to the 3d-orbital, giving it a covalency of three.

Due to the lack of unoccupied d-orbitals in the second energy shell, fluorine cannot expand its octet. As a result, it can only have one covalency.

45. Out of H_2O and H_2S , which one has higher bond angle and why?

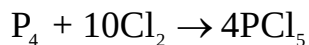
Ans: Oxygen is more electronegative than sulphur, the bond angle of H_2O is greater, and the bond pair electron of an OH bond will be closer to oxygen, causing bond-pair bond-pair repulsion between bond pairs of two OH bonds.

46. SF_6 is known but SCl_6 is not. Why?

Ans: Six F atoms can be accommodated around sulphur due to fluorine's tiny size, however the chloride ion is comparably bigger, resulting in interatomic repulsion.

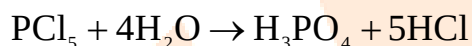
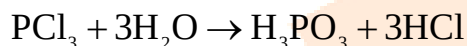
47. On reaction with Cl_2 , phosphorus forms two types of halides 'A' and 'B'. Halide A is yellowish-white powder but halide 'B' is colourless oily liquid. Identify A and B and write the formulas of their hydrolysis products.

Ans: A is PCl_5 (yellowish white powder)

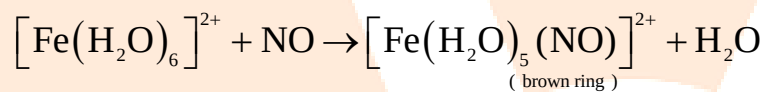
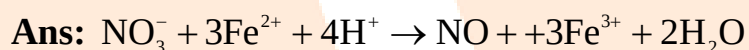


B is PCl_3 (colourless oily liquid)

Hydrolysis of the following product



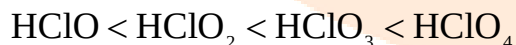
48. In the ring test of NO_3^- ion, Fe^{2+} ion reduces nitrate ion to nitric oxide, which combines with Fe^{2+} (aq) ion to form brown complex. Write the reactions involved in the formation of brown ring.



49. Explain why the stability of oxoacids of chlorine increases in the order given below: $HClO < HClO_2 < HClO_3 < HClO_4$

Ans: Oxygen is more electronegative than chlorine, the negative charge on chlorine disperses from ClO^- to ClO_4^- ion as the number of oxygen atoms linked to chlorine grows. As a result, ion stability will improve in the following order:

As the stability of the conjugate base rises, the acidic strength of the corresponding acid increases in the following order:



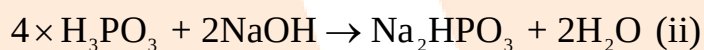
50. Explain why ozone is thermodynamically less stable than oxygen

Ans: Due its disintegration into oxygen results in the release of heat ΔH is negative) and an increase in entropy (ΔS is positive), ozone is thermodynamically unstable in comparison to oxygen. These two reactions combine to produce a substantial negative Gibbs energy change (ΔG) for its conversion to oxygen.

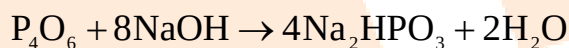
51. P_4O_6 reacts with water according to equation $P_4O_6 + 6H_2O \rightarrow 4H_3PO_3$. Calculate the volume of 0.1 M NaOH solution required to neutralise the acid formed by dissolving 1.1 g of P_4O_6 in H_2O .

Ans: $P_4O_6 + 6H_2O \rightarrow 4H_3PO_3$ (i)

Next is neutralization



On addition of 2 equations



$$P_4O_6 (\text{mol.mass}) = (4 \times 31 + 16 \times 6) = 220$$

$$\text{Number of moles of } P_4O_6 = \frac{\text{MolarMass}}{\text{Givenmass}} = \frac{1.1}{220}$$

The product formed is neutralized 8 moles of NaOH

$$P_4O_6 = 8 \times \frac{1.1}{220} = \frac{8.8}{220} \text{ mol NaOH}$$

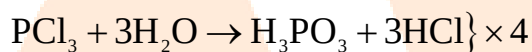
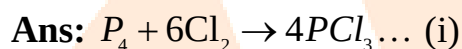
Molarity of NaOH in 1 litre = 0.1M

$$\text{Molarity} = \frac{\text{No.of Moles}}{\text{Volume in litres}}$$

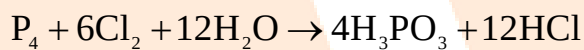
$$\text{Volume} = \frac{\text{No.of Moles}}{\text{Molarity}}$$

$$= \frac{8.8}{220} \times \frac{1}{0.1} = 0.4 \text{ L}$$

52. White phosphorus reacts with chlorine and the product hydrolyses in the presence of water. Calculate the mass of HCl obtained by the hydrolysis of the product formed by the reaction of 62 g of white phosphorus with chlorine in the presence of water



On adding eq. (i) and (ii)



1mol of white phosphorus produces 12mol of HCl

62g of white phosphorus has been taken which is equivalent to $\frac{62}{124} = \frac{1}{2}$ mol.

Therefore 6molHCl will be formed.

$$\text{Mass of 6molHCl} = 6 \times 36.5 = 219.0 \text{ gHCl}$$

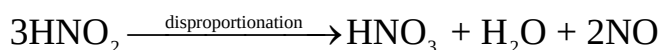
53. Name three oxoacids of nitrogen. Write the disproportionation reaction of that oxoacid of nitrogen in which nitrogen is in +3 oxidation state.

Ans: Three oxoacids are

(a) HNO_2 , Nitrous acid

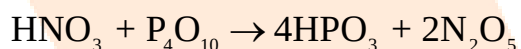
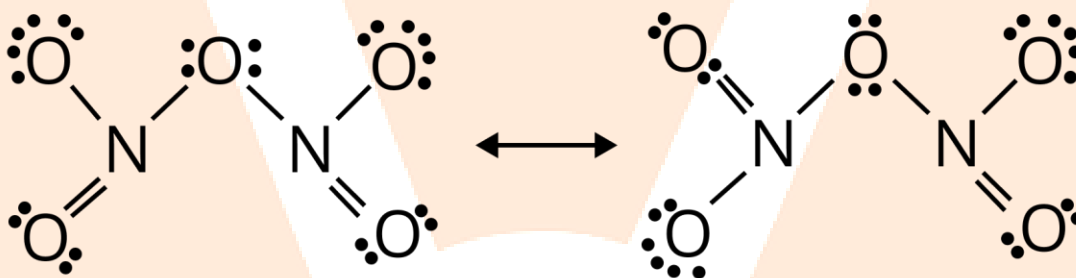
(b) HNO_3 , Nitric acid

(c) hyponitrous acid, $H_2N_2O_2$



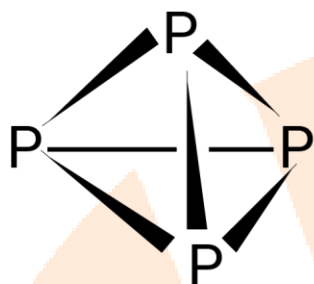
54. Nitric acid forms an oxide of nitrogen on reaction with P_4O_{10} . Write the reaction involved. Also write the resonating structures of the oxide of nitrogen formed.

Ans: On interaction with P_4O_{10} forms an oxide of nitrogen, N_2O_5 , and metaphosphoric acid, 3, nitric acid is formed. HPO_3

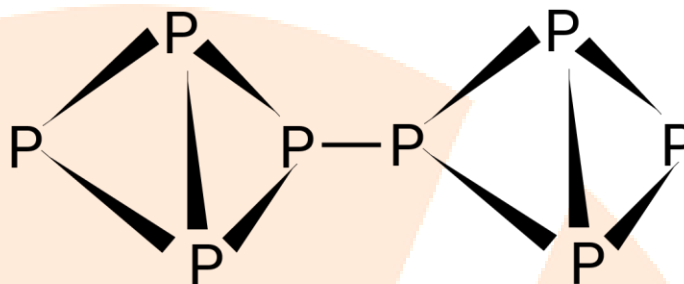


55. Phosphorus has three allotropic forms — (i) white phosphorus (ii) red phosphorus and (iii) black phosphorus. Write the difference between white and red phosphorus since their structure and reactivity.

Ans: The crystal structure of red phosphorus features a sophisticated network of bonding, whereas white phosphorus is made up of P_4 molecules. To avoid natural combustion, white phosphorus must be stored in water, but red phosphorus is stable in air.



White Phosphorous



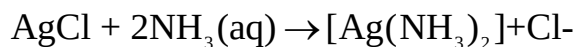
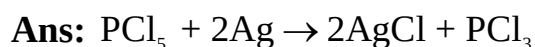
Red Phosphorous

56. Give an example to show the effect of concentration of nitric acid on the formation of oxidation product.

Ans: When nitric acid is mixed with copper metal, it produces distinct oxidation products.

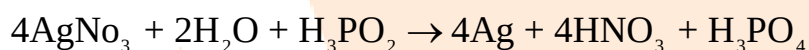


57. PCl_5 reacts with finely divided silver on heating and a white silver salt is obtained, which dissolves on adding excess aqueous NH_3 solution. Write the reactions involved to explain what happens.



58. Phosphorus forms several oxoacids. Out of these oxoacids phosphinic acid has strong reducing property. Write its structure and write a reaction showing its reducing behaviour

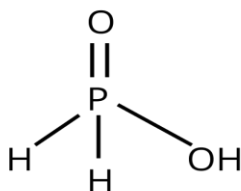
Ans: The following reaction with silver nitrate demonstrates phosphinic acid's reducing behaviour:



IV. Matching Type

Note Match the items of Column I and Column II in the following questions.

59. Match the compounds given in Column I with the hybridisation and shape given in Column II and mark the correct option



Column I	Column II
(A) XeF_6	(1) sp^3d^3 – distorted octahedral
(B) XeO_3	(2) sp^3d^2 - square planar
(C) XeOF_4	(3) sp^3 - pyramidal
(D) XeF_4	(4) sp^3d^2 - square pyramidal

Code

(i) A (1)	B (3)	C (4)	D (2)
(ii) A (1)	B (2)	C (4)	D (3)
(iii) A (4)	B (3)	C (1)	D (2)
(iv) A (4)	B (1)	C (2)	D (3)

Ans: Correct option is (i)

In XeF_6 , the central Xe atom undergoes $sp^3 d^3$ hybridisation and the molecular geometry is distorted octahedral with 1 lone pair and 6 bond pairs of electrons.

In XeO_3 , the central Xe atom undergoes sp^3 hybridisation and the molecular geometry is pyramidal with 1 lone pair and 3 bonding domains of electrons.

In XeOF_4 , the central Xe atom undergoes $sp^3 d^2$ hybridization and the molecular geometry is square pyramidal with 1 lone pair and 5 bond pairs of electrons.

In XeF_4 , the central Xe atom undergoes $sp^3 d^2$ hybridization and the molecular geometry is square planar with 2 lone pairs and 4 bond pairs of electrons.

60. Match the formulas of oxides given in Column I with the type of oxide given in Column II and mark the correct option.

Column I	Column II
(A) Pb_3O_4	(1) Neutral oxide
(B) N_2O	(2) Acidic oxide
(C) Mn_2O_7	(3) Basic oxide
(D) Bi_2O_3	(4) Mixed oxide

Code

(i) A (1)	B (2)	C (3)	D (4)
(ii) A (4)	B (1)	C (2)	D (3)
(iii) A (3)	B (2)	C (4)	D (1)
(iv) A (4)	B (3)	C (1)	D (2)

Ans: Correct option is (ii)

- A. Pb_3O_4 is a mixed ore
 B. N_2O is a neutral oxide
 C. Mn_2O_7 is acidic oxide
 D. Bi_2O_3 is basic oxide

61. Match the items of Columns I and II and mark the correct option.

Column I	Column II
(A) H_2SO_4	(1) Highest electron gain enthalpy
(B) CCl_3NO_2	(2) Chalcogen
(C) Cl_2	(3) Tear gas
(D) Sulphur	(4) Storage batteries

Code:

(i) A (4)	B (3)	C (1)	D (2)
(ii) A (3)	B (4)	C (1)	D (2)
(iii) A (4)	B (1)	C (2)	D (3)
(iv) A (2)	B (1)	C (3)	D (4)
(i) A (4)	B (3)	C (1)	D (2)
(ii) A (3)	B (4)	C (1)	D (2)

(iii) A (4)	B (1)	C (2)	D (3)
(iv) A (2)	B (1)	C (3)	D (4)

Ans: Correct option is (i)

CCl_3NO_2 stands for chloropicrin, which is known as tear gas. Cl is the element of highest electron gain enthalpy. Sulphur is the element belonging to 16th group, also known as chalcogens.

62. Match the species given in Column I with the shape given in Column II and mark the correct option.

Column I	Column II
(A) SF_4	(1) Tetrahedral
(B) BrF_3	(2) Pyramidal
(C) BrO_3^-	(3) Sea-saw shaped
(D) NH_4^+	(4) Bent T-shaped

Code:

(i) A (3)	B (2)	C (1)	D (4)
(ii) A (3)	B (4)	C (2)	D (1)
(iii) A (1)	B (2)	C (3)	D (4)
(iv) A (1)	B (4)	C (3)	D (2)

Ans: Correct option is (ii)

63. Match the items of Columns I and II and mark the correct option.

Column I	Column II
----------	-----------

(A) Its partial hydrolysis does not change oxidation state of central atom	(1) He
(B) It is used in modern diving apparatus	(2) XeF ₆
(C) It is used to provide inert atmosphere for filling electrical bulbs	(3) XeF ₄
(D) Its central atom is in sp^3d^2 hybridisation	(4) Ar

Code:

(i) A (1)	B (4)	C (2)	D (3)
(ii) A (1)	B (2)	C (3)	D (4)
(iii) A (2)	B (1)	C (4)	D (3)
(iv) A (1)	B (3)	C (2)	D (4)

Ans: Correct option is (iii)

V. Assertion and Reason Type

Note In the following questions a statement of assertion followed by a statement of reason is given. Choose the correct Answer out of the following choices.

- (i) Both assertion and reason are correct statements, and reason is the correct explanation of the assertion.
- (ii) Both assertion and reason are correct statements, but reason is not the correct explanation of the assertion.
- (iii) Assertion is correct, but reason is wrong statement.
- (iv) Assertion is wrong but reason is correct statement.
- (v) Both assertion and reason are wrong statements.

64. Assertion N₂ is less reactive than P₄.

Reason Nitrogen has more electron gain enthalpy than phosphorus.

Ans: Correct option is (iii)

Nitrogen gas has a complete octet structure for both atoms and is unreactive due to the presence of a strong triple bond. On the other hand, phosphorus has bonds with unstable angles strains compared to nitrogen, therefore it burns quickly, thus readily reacts. Nitrogen has a lower electron gain enthalpy than phosphorus.

65. Assertion HNO_3 makes iron passive.

Reason HNO_3 forms a protective layer of ferric nitrate on the surface of iron.

Ans: Correct option is (iii)

HNO_3 forms an oxide layer on the surface of iron. This is also known as corrosion or rusting.

66. Assertion HI cannot be prepared by the reaction of KI with concentrated H_2SO_4

Reason HI has lowest H–X bond strength among halogen acids

Ans: Correct option is (ii)

Because H_2SO_4 is a stronger oxidising agent and HI is a stronger reducing agent, H_2SO_4 oxidizes and is reduced to SO_2 .

67. Assertion Both rhombic and monoclinic sulphur exist as S_8 but oxygen exists as O_2 .

Reason Oxygen forms $p\pi - p\pi$ multiple bond due to small size and small bond length but $p\pi - p\pi$ bonding is not possible in sulphur.

Ans: Correct option is (i)

The rhombic and the monoclinic forms of sulphur exist as S_8 . Oxygen forms a $p\pi-p\pi$ multiple bond due to its small size and short bond length, but $p\pi-p\pi$ bonding is not possible because of sulphur's larger atomic size.

68. Assertion NaCl reacts with concentrated H_2SO_4 to give colourless fumes with pungent smell. But on adding MnO_2 the fumes become greenish yellow.

Reason MnO_2 oxidises HCl to chlorine gas which is greenish yellow.

Ans: Correct option is (i)

When NaCl reacts with concentrated H_2SO_4 , it produces colourless fumes with a pungent smell, but when MnO_2 is added, the fumes turn greenish yellow. MnO_2 oxidizes HCl to chlorine gas, which then turns greenish yellow.

69. Assertion SF_6 cannot be hydrolysed but SF_4 can be.

Reason Six F atoms in SF_6 prevent the attack of H_2O on sulphur atom of SF_6

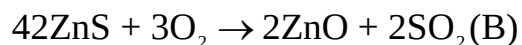
Ans: Correct option is (i)

Here, the S atom is surrounded by 6 F atoms, so it's completely occupied for H_2O to attack SF_6 , but in case of SF_4 , it's surrounded by 4F atoms, so it's to attack H_2O .

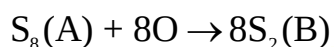
VI. Long Answer Type Question

70. An amorphous solid "A" burns in air to form a gas "B" which turns lime water milky. The gas is also produced as a by-product during roasting of sulphide ore. This gas decolourises acidified aqueous KMnO_4 solution and reduces Fe^{3+} to Fe^{2+} . Identify the solid "A" and the gas "B" and write the reactions involved.

Ans: Gas 'B' is produced as a by-product during the roasting of sulphide, so gas 'B' must be SO_2 .



Since gas 'B' is formed when amorphous solid 'A' burns in air, amorphous solid 'A' must be sulphur, S_8 .



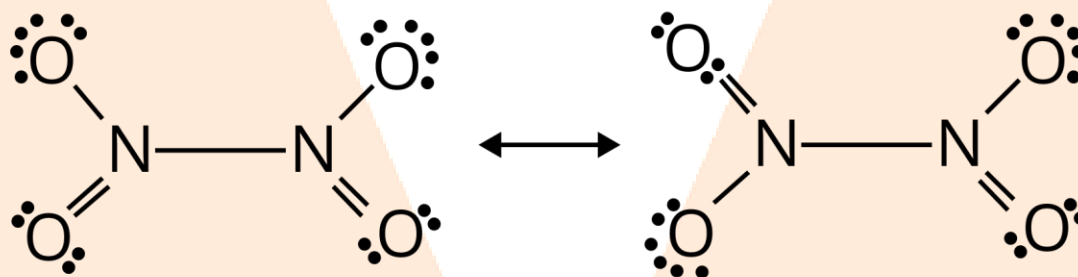
71. On heating lead (II) nitrate gives a brown gas “A”. The gas “A” on cooling changes to colourless solid “B”. Solid “B” on heating with NO changes to a blue solid ‘C’. Identify ‘A’, ‘B’ and ‘C’ and write reactions involved and draw the structures of ‘B’ and ‘C’.

Ans: Here, ‘A’ is NO_2 (Nitrogen dioxide)

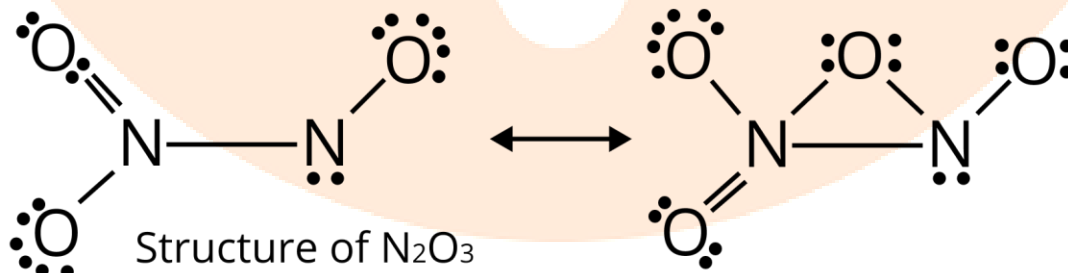
‘B’ is N_2O_4 (dinitrogen tetraoxide)

‘C’ is N_2O_3 (dinitrogen trioxide)

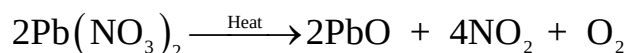
A brown gas is produced when lead nitrate(II) is heated

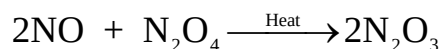
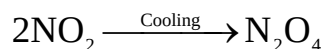


Structure of N_2O_4



Structure of N_2O_3





72. On heating compound (A) gives a gas (B) which is a constituent of air. This gas when treated with 3 mol of hydrogen (H_2) in the presence of a catalyst gives another gas (C) which is basic in nature. Gas C on further oxidation in moist condition gives a compound (D) which is a part of acid rain. Identify compounds (A) to (D) also give necessary equations of all the steps involved.

Ans: A = NH_4NO_2

B = N_2

C = NH_3

D = HNO_3

